NextTech Limited

SRS FOR CARTNEXUS

CartNexus - Men’s FashionE-Commerce Platform

Internship Title: E-Commerce Website
Batch: 26/220

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Document	Details
Version	1.0
Date	2026-04-09
Project	CartNexus
Stack	React (Vite), Node.js (Express), MySQL

1. Introduction

1.1 Purpose

This SRS defines functional and non-functional requirements, system context, data structures, and high-level design views (DFD levels 0–2, ERD, class diagram) for **CartNexus**, a bilingual (English / Bangla) online storefront with customer shopping flows, checkout, customer accounts, and an administrative back office.

1.2 Scope

In scope

- Product catalog discovery (categories, brands, search, filters).
- Product detail pages with optional color variants.
- Shopping cart (client-side state) and guest / registered checkout with COD-style payment recording.
- Customer registration, login, profile, saved addresses, and order history.
- Admin: dashboard analytics, catalog (categories, brands, products), orders, inventory, users, home hero, blog, CMS pages (terms/privacy/faqs), store settings (contact, social, WhatsApp/Messenger).
- Public APIs and JSON responses; file uploads for avatars, catalog covers, hero images.
- Real-time admin dashboard refresh via WebSocket after new orders.

Out of scope (current codebase)

- Integrated payment gateways (card/mobile wallet) — checkout records `payment_method`; external PSP integration is future work.
- Native mobile apps (responsive web only).
- Multi-tenant / multiple stores in one deployment.

1.3 Definitions & Abbreviations

Term	Meaning
SRS	Software Requirements Specification
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
CMS	Content Management (editable legal/info pages)
COD	Cash on delivery
JWT	JSON Web Token (auth)

1.4 References

- Canonical DB: `backend/db/schema.sql`
- Change log: `backend/db/db-changelog.txt`

2. Overall Description

2.1 Product Perspective

CartNexus is a **three-tier** application:

1. **Presentation:** React SPA (`frontend/`), i18n (EN/BN).
2. **Application:** REST API + WebSocket (`backend/src/server.js`, routes under `/api`).
3. **Data:** MySQL (`cartnexus` database).

External actors: **Visitors/Customers**, **Administrators**. Optional: email/SMS providers (not hard-required in core).

2.2 User Classes

User class	Description
Guest	Browse, add to cart, checkout without account (or with optional account).
Customer	Registered user: profile, addresses, orders.
Admin	Full back-office access via JWT-protected `/api/admin/`*.

2.3 Operating Constraints

- Node.js LTS, modern evergreen browsers.
- MySQL 8+ recommended; UTF-8 (`utf8mb4`).
- Environment variables for DB credentials, JWT secret, optional `VITE_API_URL`.

3. Functional Requirements

3.1 Storefront (FR-SF)

ID	Requirement
FR-SF-01	Display home hero from `home_hero` (API).
FR-SF-02	List/filter products by category, brand, price, stock; product detail with variants when configured.
FR-SF-03	Shopping cart persisted in browser (context); quantities and variant selection.

ID	Requirement
FR-SF-04	Checkout collects delivery data; creates order via `POST /api/orders`; supports guest or Bearer token.
FR-SF-05	Static/marketing pages: About, Contact (store settings), Blog, Terms/Privacy/FAQs (CMS API when available).
FR-SF-06	Footer/store settings: social links, WhatsApp/Messenger FAB when configured.

3.2 Customer Account (FR-CU)

ID	Requirement
FR-CU-01	Register, login, JWT session (storage).
FR-CU-02	Profile (name, phone, avatar upload).
FR-CU-03	CRUD addresses (`user_addresses`).
FR-CU-04	List own orders (`GET /api/orders` with customer token).

3.3 Administration (FR-AD)

ID	Requirement
FR-AD-01	JWT login; role `admin` required for `/api/admin/`*.
FR-AD-02	Dashboard: revenue, orders, charts, tasks, suggestions (API-driven).
FR-AD-03	Manage categories, brands, products (including descriptions, variants, covers).
FR-AD-04	Manage orders (status updates); inventory adjustments and movement history.
FR-AD-05	Manage users (list, role); blog posts; CMS HTML pages; home hero; store settings.
FR-AD-06	WebSocket subscription for dashboard refresh on relevant events.

3.4 Data & Integration (FR-DT)

ID	Requirement
FR-DT-01	Public read APIs for catalog, blog, CMS pages, store settings, home hero.
FR-DT-02	Uploaded files served under `/uploads/`*.
FR-DT-03	Orders immutable line items snapshot in `order_items`.

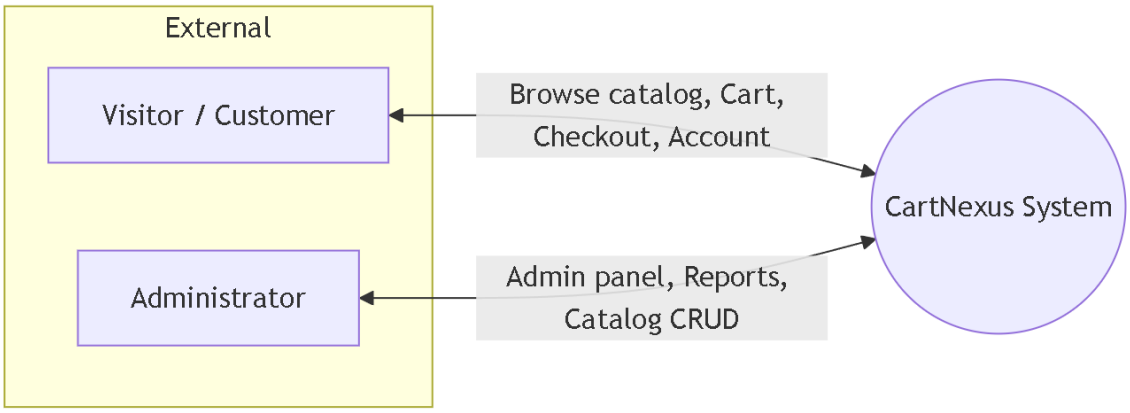
4. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Catalog and product APIs are paginated/filtered; avoid unbounded scans in admin lists.
NFR-02	Security	Password hashing (bcrypt); admin/customer JWT separation; HTTPS in production.
NFR-03	Availability	Graceful degradation if optional tables/features are missing (503 with coded errors where implemented).
NFR-04	i18n	UI strings EN/BN; product/content bilingual fields were modeled.
NFR-05	Maintainability	Migrations + changelog for schema drift.

5. Data Flow Diagrams (DFD)

5.1 Level 0 (Context Diagram)

Shows the system boundary and external interactions.

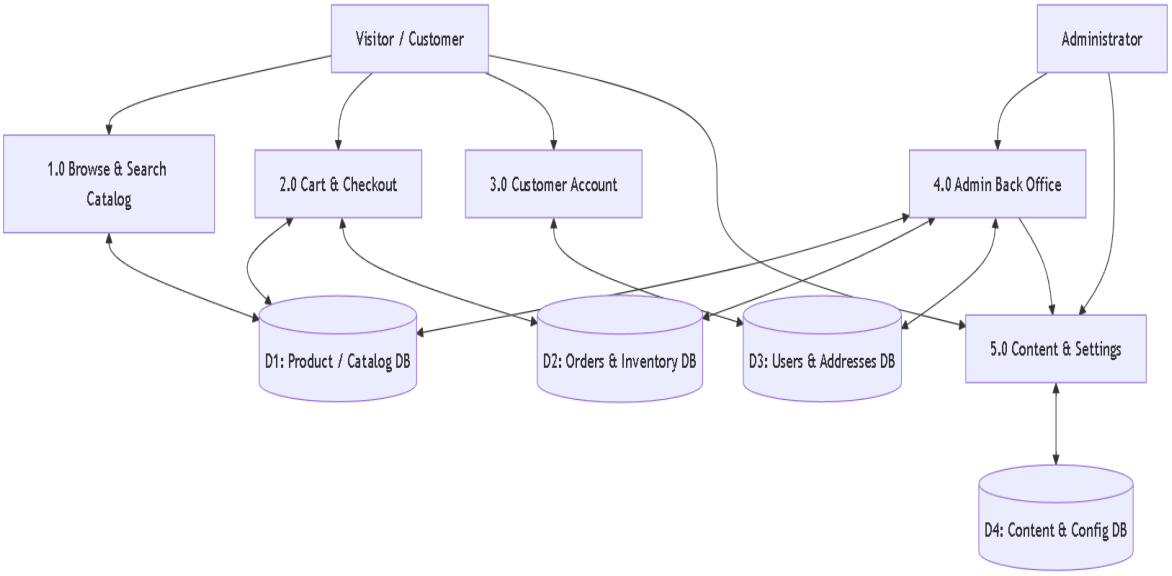


Data flows (summary)

- **Customer → System:** search/browse requests, cart actions, checkout payload, login credentials, profile updates.
- **System → Customer:** product/catalog data, order confirmation, auth tokens.
- **Admin → System:** CRUD operations, file uploads, configuration.
- **System → Admin:** analytics, lists, confirmations

5.2 Level 1 DFD

Major processes inside CartNexus and shared data stores.

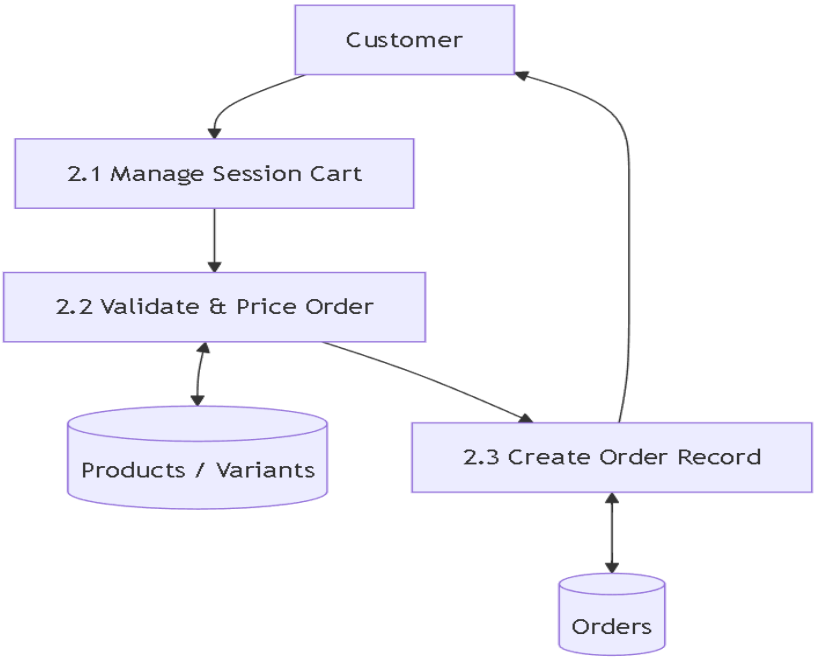


Store grouping (logical)

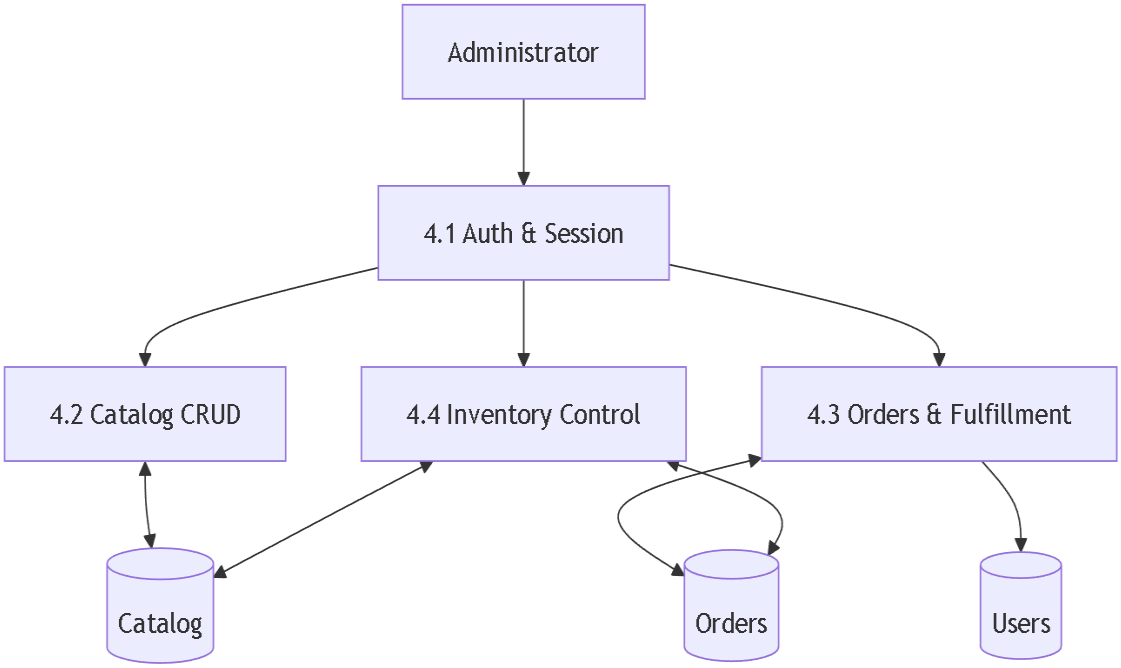
Store	Typical tables
D1	`categories`, `brands`, `products`, `product_color_variants`
D2	`orders`, `order_items`, `inventory_stock_movements`
D3	`users`, `user_addresses`, `admin_task_completions`
D4	`home_hero`, `blog_posts`, `cms_pages`, `store_settings`

5.3 Level 2 DFD (decomposition)

5.3.1 Process 2.0 — Cart & Checkout (expanded)

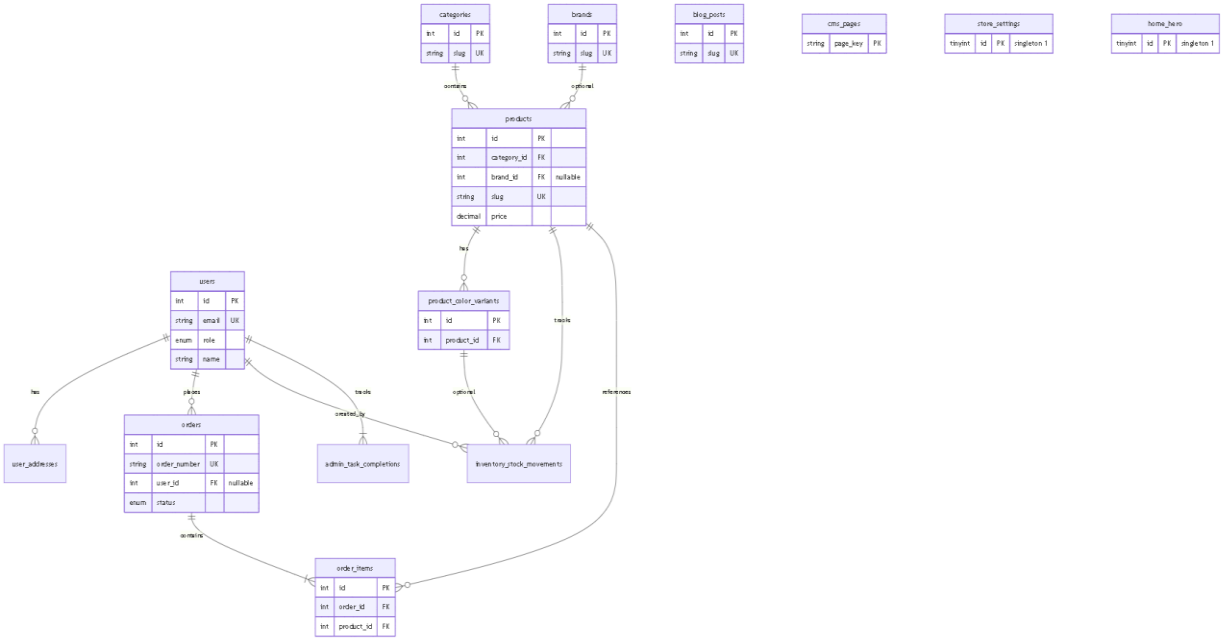


5.3.2 Process 4.0 — Admin Back Office (expanded)



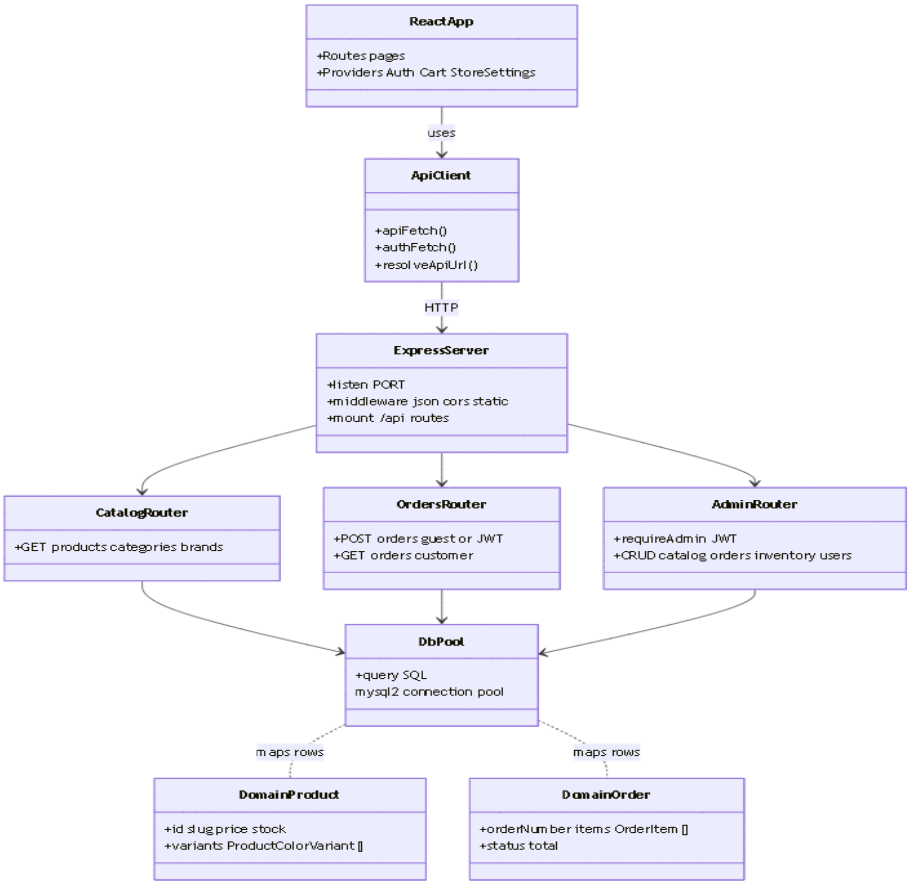
6. Entity Relationship Diagram (ERD)

Core relationships are aligned with `backend/db/schema.sql`. Cardinality in notes.



7. Class Diagram (Logical / Layered)

The codebase is primarily **JavaScript modules** (not Java-style classes everywhere). The diagram below captures **logical responsibilities** (UML-style) mapped to major folders.



8. Budget (Indicative)

Figures are **planning estimates** in **BDT** (Bangladesh Taka). Adjust for team rates, timeline, and hosting choice.

Item	Description	Low (BDT)	High (BDT)
A. One-time development	Design + storefront + admin + DB + deploy (already largely implemented; use for future phases or maintenance)	200,000	800,000+
B. Domain & SSL	`.com` / `.bd` domain + TLS (often included with hosting)	1,500	5,000 / yr
C. Hosting	VPS (2–4 GB RAM) or cloud VM for Node + MySQL	12,000	60,000 / yr
D. Database	Same VM or managed MySQL add-on	0	30,000 / yr
E. Storage & CDN	Product images on server or object storage	0	15,000 / yr
F. Email / SMS	Transactional email; optional OTP/SMS	0	24,000 / yr
G. Backup & monitoring	Automated DB backup, uptime	0	12,000 / yr
H. Maintenance	Small fixes, dependency updates, minor features (optional retainer)	30,000	120,000 / yr

Rough first-year total (excluding large new features): ~55,000 – 250,000 BDT recurring + variable dev.

9. Revision History

Version	Date	Notes
1.0	2026-04-09	Initial SRS with DFD 0–2, ERD, class diagram, budget